

Association of the Extent of ischemic Core and Penumbra with the Benefits of Endovascular Thrombectomy: Analysis of the SELECT2 Trial

Statistical Analysis Protocol

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Ischemic injury estimates on different imaging modalities and their association with outcomes and thrombectomy treatment effect in patients with large ischemic core strokes

– A pre-specified secondary analysis of SELECT2 trial

Statistical Analysis Protocol

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1. Introduction

This pre-specified secondary analysis describes key imaging characteristics and evaluates their association with functional and safety outcomes and thrombectomy treatment effect in patients with large ischemic core strokes enrolled in SELECT2 trial.

2. Primary Study Aim

The aim of the study is to evaluate the association of key imaging characteristics, namely 1) CT ASPECTS, 2) Ischemic core estimates on CT perfusion/MRI Diffusion and 3) mismatch on CT/MR perfusion; with functional and safety outcomes and thrombectomy treatment effect in patients with large ischemic core strokes enrolled in SELECT2 trial.

3. Study population

All patients enrolled in SELECT2 trial will be included in the analysis for thrombectomy treatment effect based on various imaging characteristics and profiles.

4. Imaging definitions

- Core lab evaluated CT ASPECTS will be used for all analyses.
- Ischemic volume on CT/MR perfusion imaging will be defined using rCBF <30% volumes from CT perfusion/ADC <620 X 10⁶mm/s² from MR perfusion/diffusion using automated processing by RAPID software.
- CT hypodensity volume using manual delineation will be calculated for all baseline non-contrast CT images.
- Primary analysis will be completed using larger of the two ischemic volume estimates (1) rCBF <30% volume for CT perfusion or ADC <620 X 10⁶mm/s² from MR perfusion/diffusion and 2) CT hypodensity volume.
- A sensitivity analysis using rCBF <30% volume for CT perfusion or ADC <620 X 10⁶mm/s² from MR perfusion/diffusion as ischemic core volume will also be completed.
- Mismatch volume will be determined as volumetric difference between critically hypoperfused tissue (Tmax >6s) volume and ischemic core volume. Mismatch ratio will be determined as a ratio of ischemic core volume and critically hypoperfused tissue (Tmax >6s) volume.
- Favorable mismatch profile 1 is prospectively defined as mismatch ratio of ≥1.8 & mismatch volume of ≥15ml. Favorable Mismatch profile 2 is prospectively defined as mismatch ratio of ≥1.2 & mismatch volume of ≥10ml.
- Discordant imaging profile 1 is defined as ischemic core volume <70ml and CT ASPECTS of ≤5. Discordant imaging profile 2 is defined as ischemic core volume ≥70ml and CT ASPECTS of >5. Concordant imaging profile is defined as ischemic core volume ≥70ml and CT ASPECTS of ≤5.

5. Statistical analysis

5.1. Principles

- All analyses will be performed under an “as-treated” approach.
- All analyses will be exploratory in nature and no adjustment for multiple comparison will be performed.

- Continuous variables will be included in regression models if no substantial deviation in linearity assumption is found. In cases with substantial deviation from linearity assumption, appropriate transformations of variables will be considered.
- Predictive margins and marginal probabilities will be used to graphically illustrate how imaging characteristics affect the outcomes and thrombectomy treatment effect.
- Patients with mRS scores of 6 (death) will be merged with patients who achieved scores of 5 (bedridden with constant care required) to avoid counting a shift from mRS 6 to 5 as a treatment success, for analyses evaluating distribution of modified Rankin Scale scores.
- Analyses will be conducted primarily using Stata and R statistical software.
- Missing data will not be imputed.

5.2. Outcomes definitions and analysis approach:

1) Distribution of modified Rankin Scale scores at 90-day follow-up:

- Descriptive statistics including tabulation and stacked bar graphs may be used to illustrate the distribution of modified Rankin Scale scores across various categories of independent variable under consideration.
- The association of a given imaging characteristics with distribution of modified Rankin Scale scores will be evaluated using probabilistic index model, adjusting for age, NIHSS at presentation and time from last known well to randomization and ischemic core volume as prognostic variables. Modified Rankin Scale scores of 5 and 6 will be merged to avoid counting a shift from mRS 6 to 5 as a treatment success, in consistency with the primary manuscript and analysis approach.
- For subgroup analyses of one of these prognostic variables, that variable will be excluded from adjustment (e.g. analyses stratified by core volume will not adjust for core volume).

2) Proportion of patients achieving functional independence (mRS 0-2):

- Descriptive statistics including tabulation and bar graphs may be used to illustrate the proportion of patients achieving functional independence across various categories of independent variable under consideration.
- The association of a given imaging characteristics with functional independence (mRS 0-2) will be evaluated using a modified Poisson regression model with robust standard errors, adjusting for age, NIHSS at presentation, time from last known well to randomization and ischemic core volume as prognostic variables.
- If the calibration of predicted probabilities is poor for the Poisson regression model and logistic regression produces better calibration, logistic regression will be used instead for the purposes of producing marginal probability plots.
- For subgroup analyses of one of these prognostic variables, that variable will be excluded from adjustment (e.g. analyses stratified by core volume will not adjust for core volume)

3) Proportion of patients achieving independent ambulation (mRS 0-3):

- Descriptive statistics including tabulation and bar graphs may will be used to illustrate the proportion of patients with independent ambulation across various categories of independent variable under consideration.
- The association of a given imaging characteristics with independent ambulation (mRS 0-3) will be evaluated using a modified Poisson regression model with robust standard

errors, adjusting for age, NIHSS at presentation, time from last known well to randomization and ischemic core volume as prognostic variables.

- For subgroup analyses of one of these prognostic variables, that variable will be excluded from adjustment (e.g. analyses stratified by core volume will not adjust for core volume)
- If the calibration of predicted probabilities is poor for the Poisson regression model and logistic regression produces better calibration, logistic regression will be used instead for the purposes of producing marginal probability plots.
- In addition to the marginal predicted probabilities reported for other outcome measures, predicted probabilities of achieving independent ambulation based on the combination of age and time from last known well to randomization within the subgroup of patients who received EVT.

4) Proportion of patients experiencing complete dependence or death (mRS 5-6):

- Descriptive statistics including tabulation and bar graphs may will be used to illustrate the proportion of patients with independent ambulation across various categories of independent variable under consideration.
- The association of a given imaging characteristics with death or severe disability (mRS 5-6) will be evaluated using a modified Poisson regression model with robust standard errors, adjusting for age, NIHSS at presentation, time from last known well to randomization and ischemic core volume as prognostic variables.
- If the calibration of predicted probabilities is poor for the Poisson regression model and logistic regression produces better calibration, logistic regression will be used instead for the purposes of producing marginal probability plots.
- For subgroup analyses of one of these prognostic variables, that variable will be excluded from adjustment (e.g. analyses stratified by core volume will not adjust for core volume)

5) Proportion of patients experiencing symptomatic intracerebral hemorrhage:

- Symptomatic intracerebral hemorrhage is defined as parenchymal hemorrhage type II or remote parenchymal hemorrhage on follow-up imaging associated with a 4-point or larger increase in NIHSS score at 4-hour follow-up window.
- Due to very few events observed in the study, number & proportion of patients experiencing symptomatic intracerebral hemorrhage will be reported but no further analysis or statistical testing or modelling will be performed.

5.3. Analyses to be performed:

5.3.1 *Effect of ischemic core volume on outcomes and treatment effect*

- Patients will be stratified using ischemic core thresholds of 70ml, 100ml and 150ml. (< 70mL vs $\geq 70\text{mL}$; < 100mL vs $\geq 100\text{mL}$; <150mL vs $\geq 150\text{mL}$)
- Evaluation of outcomes and thrombectomy treatment effect will be completed as described in section 5.2.
- Interaction terms between ischemic core volume and treatment will be used to illustrate any heterogeneity of treatment effect.

5.3.2 *Effect of ASPECTS on non-contrast CT on outcomes and treatment effect*

- Patients will be stratified using ASPECTS thresholds of 2 and 5 (0-2, 3-5, 6-10).

- Evaluation of outcomes and thrombectomy treatment effect will be completed as described in section 5.2.
- Additionally, within patients with ASPECTS 3-5, we will further stratify patients at each ASPECTS score point and repeat analyses described in section 5.2.
- Interaction terms between ASPECTS and treatment will be used to illustrate any heterogeneity of treatment effect.

5.3.3 Effect of mismatch on perfusion imaging

- Patients will be stratified using pre-specified mismatch profiles described in section 4.
- Additionally, patients will be stratified based on mismatch volume thresholds of 50ml & 100ml.
- Evaluation of outcomes and thrombectomy treatment effect will be completed as described in section 5.2.
- Interaction terms between a given mismatch profile and treatment will be used to illustrate any heterogeneity of treatment effect.
- Additionally, the association of mismatch volume with EVT outcomes will also be evaluated, considering mismatch volume as a continuous variable.

5.3.4 Effect of concordance and discordance of imaging profiles

- Patients will be stratified using pre-specified concordant/discordant imaging profiles described in section 4.
- Evaluation of outcomes and thrombectomy treatment effect will be completed as described in section 5.2.
- Interaction terms between imaging profiles and treatment will be used to illustrate any heterogeneity of treatment effect.

Appendix I: Appendix to the Statistical Analysis Protocol

Appendix to the Statistical Analysis Protocol for Ischemic injury estimates on different imaging modalities and their association with outcomes and thrombectomy treatment effect in patients with large ischemic core strokes – A pre-specified secondary analysis of SELECT2 trial

The following analyses were included in a post-hoc manner after statistical analysis plan was finalized to further examine and characterize the association of different imaging characteristics with clinical outcomes and EVT treatment effect.

1. Evaluation of CT hypodensity volumes as third volumetric estimate of ischemic changes in addition to pre-specified composite core (larger of CTP/MRI core and CT hypodensity) and CTP/MRI core volume
2. Evaluation of mismatch using larger of the two modalities
3. Evaluation of patients with significant overestimation of infarct core
4. Evaluation of follow-up infarct volume
5. Evaluation of within-patient CT hypodensity and CTP/MRI core